

Full Stack Development Summary

- Functions
- Types of Functions
- Return Keyword
- Components of a Class
- Types of Variables and its uses.
- What is Constructor
- Types Constructor
- “this” keyword
- Static Block & Instance Block

FUNCTION

- A function is called method in java is a block of code that performs a special task.
- It helps make your code reusable, organized, and easier to read.

COMPONENTS

- type—data type of value the method returns(or void if it returns nothing).
- Method name—name of the method(follows camel case naming).
- Parameters—optional input values(like variable passed to the method).
- Method body—the code that runs when the method is call.

TYPES OF FUNCTIONS

- Function with no parameters and no return value.
- Function with parameters and no return value .
- Function with no parameter and a return value.
- Function with parameters and a return value.

RETURN KEYWORD

- If the method 's return type is not void you must return a value that matches the method's declared datatype.
- If the method's return type is void, you can use return, (optional) just to exit the method early.
- Exit the method and send a value back to the caller if the method has a return type.

COMPONENTS OF A CLASS

- A Class in java is a blueprint that defines how an object behave and values and what data it holds.
- Variable (data members)-store data.
- Methods (member functions)-define actions.
- Constructor—initialize objects.
- Blocks (static/instance)-execute specific code.
- Nested class- inner class (optional).

TYPES OF VARIABLES AND USES

- Instance variable = employee name (different for each employee).
- Static variable = company name (same for all).
- Local variable = temporary data like bonus calculations.

WHAT IS CONSTRUCTOR

- A Constructor is a special method used to create (**initialize objects in java**).
- Its name is always the same as the class.
- It does not have a return type(**not even void**).
- It is automatically called when you create an object using new.

TYPES OF CONSTRUCTOR

- Default constructor-no parameters,assigns default values.
- Parameterized constructor-takes parameters to assign default values.
- Copy constructor-copies value from another object.

THIS KEYWORD

- A Method need to access to the entire object,not just obe variable.
- We want to send the current object to another class or method.
- We want object to object interaction.
- This refers to the current object of the class.
- It is used when local variables and instance variables have the same name.

STATIC BLOCK

- **A Static block is executed once when the class is loaded before the main() method.**
- Its mainly used to initialize static variables.

INSTANCE BLOCK

- An instance block is executed every time an object is created, before the constructor.
- Runs every time an object is created
- Declared as {} without static.